

Formal MIP Formulation of the Car Sequencing Problem

Sets

I set of car types, $|I| = n$,
 J set of sequence positions, $|J| = m$,
 K set of colors, $|K| = o$.

Parameters

A_i availability of car type $i \in I$,
 $\text{color}(i) \in K$ color of car type i .

Decision Variables

$$x_{i,j} \in \{0, 1\} \quad \forall i \in I, j \in J,$$

where $x_{i,j} = 1$ iff car type i is assigned to position j .

Core Constraints

Assignment

$$\sum_{i \in I} x_{i,j} = 1 \quad \forall j \in J.$$

Availability

$$\sum_{j \in J} x_{i,j} \leq A_i \quad \forall i \in I.$$

Working Constraints

Sliding Window Capacity (SLWINCAP)

Given $S \subseteq I$, window length p and capacity q :

$$\sum_{t=j}^{j+p-1} \sum_{i \in S} x_{i,t} \leq q \quad \forall j = 1, \dots, m - p + 1.$$

Special cases:

$$\text{MAXINROW}(r) : q = r, p = r + 1, \quad \text{MINGAP}(r) : q = 1, p = r + 1.$$

NOTFOLLOW

Given $S, T \subseteq I$ and horizon p :

$$\sum_{i \in S} p x_{i,j} + \sum_{t=j+1}^{\min(j+p,m)} \sum_{r \in T} x_{r,t} \leq p \quad \forall j = 1, \dots, m-1.$$

SUM

Given $S \subseteq I$, segment $[a, b]$ and integer q :

$$\sum_{j=a}^b \sum_{i \in S} x_{i,j} \# q, \quad \# \in \{\leq, =, \geq\}.$$

PRECEDEALL

Given $S, T \subseteq I$, introduce $y \in \mathbb{R}$:

$$\begin{aligned} \sum_{i \in S} j x_{i,j} - y &\leq 0 \quad \forall j \in J, \\ \sum_{r \in T} (m - j + 1) x_{r,j} + y &\leq m \quad \forall j \in J. \end{aligned}$$

Objective: Minimization of Color Transitions

Version 0

Feasibility model without objective.

Version 1

Introduce $c_{j,k} \in \{0, 1\}$:

$$\begin{aligned} \max \sum_{j=2}^m \sum_{k \in K} c_{j,k}, \\ -2c_{j,k} + \sum_{i: \text{color}(i)=k} (x_{i,j-1} + x_{i,j}) &\geq 0. \end{aligned}$$

Version 2

Introduce $c_j \geq 0$:

$$\begin{aligned} \min \sum_{j=2}^m c_j, \\ c_j + \sum_{i: \text{color}(i)=k} (x_{i,j-1} - x_{i,j}) &\geq 0 \quad \forall k \in K. \end{aligned}$$

Version 3

Binary c_j with index-encoded colors:

$$\begin{aligned} (o-1)c_j + \sum_{k \in K} \sum_{i: \text{color}(i)=k} k(x_{i,j-1} - x_{i,j}) &\geq 0, \\ -(o-1)c_j + \sum_{k \in K} \sum_{i: \text{color}(i)=k} k(x_{i,j-1} - x_{i,j}) &\leq 0. \end{aligned}$$

Version 4

Per-color transition variables $c_{j,k} \geq 0$:

$$\begin{aligned} & \min \sum_{j=2}^m \sum_{k \in K} c_{j,k}, \\ & \sum_{i: \text{color}(i)=k} x_{i,j-1} + \sum_{h \neq k} \sum_{r: \text{color}(r)=h} x_{r,j} - c_{j,k} \leq 1. \end{aligned}$$

Version 5

Single transition variable $c_j \geq 0$:

$$\begin{aligned} & \min \sum_{j=2}^m c_j, \\ & \sum_{i: \text{color}(i)=k} x_{i,j-1} + \sum_{h \neq k} \sum_{r: \text{color}(r)=h} x_{r,j} - c_j \leq 1 \quad \forall k \in K. \end{aligned}$$